Foreword

"Change is the law of life, and those who look only to the past or present are certain to miss the future."

John F Kennedy

Intelligent systems refer to a class of software systems that use artificial intelligence (AI) techniques in order to carry out tasks which would be considered 'intelligent' if done by humans. Such systems will extend, multiply and leverage our mental and physical abilities and pervade all aspects of our lives.

The work reported in this special edition of BT Technology Journal, with the theme of 'Towards engineering intelligent systems', demonstrates the vision, breadth and depth of the intelligent systems research and development programme within BT's Applied Research and Technology department. It concentrates on AI technologies of constraint optimisation and scheduling, software agents, and soft computing.

The work reported on constraint optimisation and scheduling is not only technically innovative, but also benefiting BT immensely. For example, a paper in this issue describes the integration of AI techniques within a dynamic scheduling system central to Work Manager — BT's workforce scheduling system which schedules the activities of thousands of technicians all over the UK. It has been estimated the dynamic scheduler work would be saving BT over £140 million per annum. This work was awarded a 1997 British Computer Society Information Technology (IT) medal for business and technical contributions. Another paper reports on a deployed system which uses an AI technique to provide a solution to the problem of generating and sequencing calls to test BT's billing system as required by Oftel. The papers on software agents demonstrate the innovative nature of our agent R&D programme and our determination to develop methodological approaches for engineering agent-based systems and solutions. Our results and successes on this very young and promising AI technology are already having significant impact. For example, a paper describes BT's ZEUS tool-kit for rapidly engineering certain classes of agent applications. This work won the prestigious 1997 British Computer Society award for IT excellence and innovation, which in turn clearly reinforces BT's brand as an innovative IT company. Some of the reported work on our personal and information agents is being trialled and will be exploited soon. In addition, BT is playing a vanguard role in the standardisation of agent technology, also described in this issue. Finally, the papers on soft computing and machine intelligence clearly demonstrate our vision and pioneering longer-term research in building intelligent systems, helping to ensure a dominant position for BT in the global information society of the next millennium.

The successful downstreaming of some of the work, the national prizes and international recognition that some of the work has earned bear testimony to the unquestioned depth and quality of the innovation going on in BT. Many of the authors are already internationally recognised and I commend all of them on their efforts. I hope the reader gains some appreciation and understanding of how BT is facing up to the future through some of the work reported in this issue.

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